

For questions 1 – 6: Determine the convergence or divergence of the sequence algebraically. If the sequence converges, find the limit.

1. $\{a_n\} = \left\{ \frac{n}{2n+1} \right\}$

2. $\{a_n\} = \left\{ \frac{(-1)^{n+1} 2n^3}{3n^2 + 8n} \right\}$

3. $\{a_n\} = \left\{ \frac{(n+1)!}{(n+3)!} \right\}$

4. $\frac{1}{2}, \frac{4}{5}, \frac{9}{10}, \frac{16}{17}, \frac{25}{26}, \dots$

5. $\{a_n\} = \left\{ \frac{2^n + (-1)^n}{n^2 + 3} \right\}$

6. $\{a_n\} = \left\{ \sin\left(\frac{\pi}{n}\right) + 3 \right\}$

For questions 7 – 14: Determine whether the series converges or diverges. If it converges and the sum can be determined, state the sum.

7. $\sum_{k=1}^{\infty} \left(\frac{1}{5} \right)^{k-2}$

8. $\sum_{n=1}^{\infty} \frac{n^2 - 6n + 5}{4n^2 + 3}$

$$9. \quad \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$$

$$10. \quad \sum_{k=0}^{\infty} 4^{-k} (3^{k+1})$$

$$11. \quad \sum_{n=1}^{\infty} \frac{4n}{\sqrt[4]{n^9}}$$

$$12. \quad \sum_{n=0}^{\infty} \frac{\pi^n}{e^n}$$

$$13. \quad \frac{1}{1 \cdot 1} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 9} + \frac{1}{4 \cdot 16} + \dots$$

$$14. \quad \sum_{n=1}^{\infty} \frac{4}{\sqrt[4]{n^3}}$$

Answers

1. Since $\lim_{n \rightarrow \infty} \frac{n}{2n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2}$ by L'Hôpital, the sequence converges to $\frac{1}{2}$.
2. Ignoring the alternating part, $\lim_{n \rightarrow \infty} \frac{2n^3}{3n^2+8n} = \lim_{n \rightarrow \infty} \frac{6n^2}{6n+8} = \lim_{n \rightarrow \infty} \frac{12n}{6} = \infty$ by L'Hôpital, the sequence would alternate to $\pm \infty$, so the sequence diverges.
3. This can be simplified to $\frac{(n+1)(n)(n-1)\dots}{(n+3)(n+2)(n+1)(n)(n-1)\dots} = \frac{1}{(n+3)(n+2)}$. Then, $\lim_{n \rightarrow \infty} \frac{1}{(n+3)(n+2)} = 0$ and therefore the sequence converges to 0.
4. Look at the numerator and denominator as two different sequences to write the overall sequence. The numerators are n^2 and the denominators are all one more, or $n^2 + 1$. Therefore, the sequence is $\frac{n^2}{n^2+1}$. So, $\lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = \lim_{n \rightarrow \infty} \frac{2n}{2n} = 1$ by L'Hôpital and the sequence converges to 1.
5. Even though there is an alternating term, it is not being multiplied, so it is insignificant in comparison to the 2^n and n^2 terms. So, $\lim_{n \rightarrow \infty} \frac{2^n}{n^2+3} = \lim_{n \rightarrow \infty} \frac{2^n \ln 2}{2n} = \lim_{n \rightarrow \infty} \frac{2^n (\ln 2)^2}{2} = \infty$ by L'Hôpital, so the sequence diverges.
6. Since $\lim_{n \rightarrow \infty} \left(\sin\left(\frac{\pi}{n}\right) + 3 \right) = \sin 0 + 3 = 3$, the sequence converges to 3.
7. This is a geometric series and because $\left| \frac{1}{5} \right| < 1$, this will converge. The sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being $\left(\frac{1}{5}\right)^{1-2} = 5$. So $S = \frac{5}{1-\frac{1}{5}} = \frac{5}{\frac{4}{5}} = \frac{25}{4}$.
8. The Nth term test for divergence shows that $\lim_{n \rightarrow \infty} \frac{n^2-6n+5}{4n^2+3} = \lim_{n \rightarrow \infty} \frac{2n-6}{8n} = \lim_{n \rightarrow \infty} \frac{2}{8} = \frac{1}{4}$ and $\frac{1}{4} \neq 0$. Therefore, this series diverges.
9. This is an integral test, where $\int_2^{\infty} \frac{dn}{n(\ln n)^3}$. This requires a u-sub with $u = \ln n$, $du = \frac{dn}{n}$, so $\lim_{b \rightarrow \infty} \int_2^b \frac{du}{u^3}$. So, $\lim_{b \rightarrow \infty} \int_2^b u^{-3} du = \lim_{b \rightarrow \infty} \left. \frac{-1}{2(\ln n)^2} \right|_2^b = \lim_{b \rightarrow \infty} \left[\frac{-1}{2(\ln b)^2} - \frac{-1}{2(\ln 2)^2} \right]$. Since $\lim_{b \rightarrow \infty} \frac{-1}{2(\ln b)^2} = 0$, the series converges (note that the series does not converge to the answer to the improper integral).

10. Rewriting this shows that this is geometric, with $\sum_{k=0}^{\infty} 4^{-k} (3^{k+1}) = \sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k (3^k)(3^1) = \sum_{k=0}^{\infty} 3\left(\frac{3}{4}\right)^k$. Since

$\left|\frac{3}{4}\right| < 1$, this will converge. The sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being

$$3\left(\frac{3}{4}\right)^0 = 3. \text{ So, } S = \frac{3}{1 - \frac{3}{4}} = \frac{3}{\frac{1}{4}} = 12.$$

11. Since this has a ratio of powers, we can use p -series. $\sum_{n=1}^{\infty} \frac{4n}{\sqrt[4]{n^9}} = \sum_{n=1}^{\infty} \frac{4n}{n^{\frac{9}{4}}} = \sum_{n=1}^{\infty} \frac{4}{n^{\frac{5}{4}}}$, so $p = \frac{5}{4} > 1$, so the series converges.

12. This is geometric and can be seen as $\sum_{n=0}^{\infty} \left(\frac{\pi}{e}\right)^n$. Since $\left|\frac{\pi}{e}\right| \geq 1$, this will diverge.

13. We first must write out the series and notice that the denominator is $\frac{1}{n \cdot n^2} = \frac{1}{n^3}$, so the series is

$\sum_{n=1}^{\infty} \frac{1}{n^3}$. This is a p -series with $p = 3 > 1$, so the series converges.

14. This is a p -series with $\sum_{n=1}^{\infty} \frac{4}{n^{\frac{3}{4}}}$, so $p = \frac{3}{4} \leq 1$, so the series diverges.